Chapter 3

Binomial and Proportion Responses

Sometimes we observe more than one binary outcome for each combination of the predictors so that the response becomes more than simply binary but some number of one type and some number of the other type. In such situations, we have a binomial rather than just a Bernoulli distributed response.

3.1 Binomial Regression Model

Suppose the response variable Y_i for $i = 1,\ldots,n$ is binomially distributed $B(m_i, p_i)$ so that:

$$P(Y_i = y_i) = \binom{m_i}{y_i} p_i^{y_i}(1-p_i)^{m_i - y_i}$$

We further assume that the Y_i are independent. The individual outcomes or *trials* that compose the response Y_i are all subject to the same q predictors $(x_{i1},\ldots,x_{iq})$. The group of trials is known as a *covariate class*. For example, we might record whether customers of a particular type make a purchase or not. Conventionally, one outcome is labeled a *success* (say, making purchase in this example) and the other outcome is labeled as a *failure*. No emotional meaning should be attached to success and failure in this context. For example, success might be the label given to a patient death with survival being called a failure. Because we need to have multiple trials for each covariate class, data for binomial regression models is more likely to result from designed experiments with a few predictors at chosen values rather than observational data which is likely to be more sparse.

As in the binary case, we construct a *linear predictor*:

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_q x_{iq}$$

We can use a logistic link function $\eta_i = \log(p_i/(1-p_i))$. The log-likelihood is then given by:

$$l(\beta) = \sum_{i=1}^{n} \left[y_i \eta_i - m_i \log(1+e^{\eta_i}) + \log \binom{m_i}{y_i} \right]$$

Let's work through an example to see how the analysis differs from the binary response case.

In January 1986, the space shuttle Challenger exploded shortly after launch. An investigation was launched into the cause of the crash and attention focused on the rubber O-ring seals in the rocket boosters. At lower temperatures, rubber becomes more brittle and is a less effective sealant. At the time of the launch, the temperature

was 31°F. Could the failure of the O-rings have been predicted? In the 23 previous shuttle missions for which data exists, some evidence of damage due to blow by and erosion was recorded on some O-rings. Each shuttle had two boosters, each with three O-rings. For each mission, we know the number of O-rings out of six showing some damage and the launch temperature. This is a simplification of the problem — see Dalal et al. (1989) for more details.

We plot the proportion of damaged O-rings against temperature in Figure 3.1:

```
data(orings, package="faraway")
plot(damage/6 ~ temp, orings, xlim=c(25,85), ylim = c(0,1), xlab="
    ↪ Temperature", ylab="Prob of damage")
```

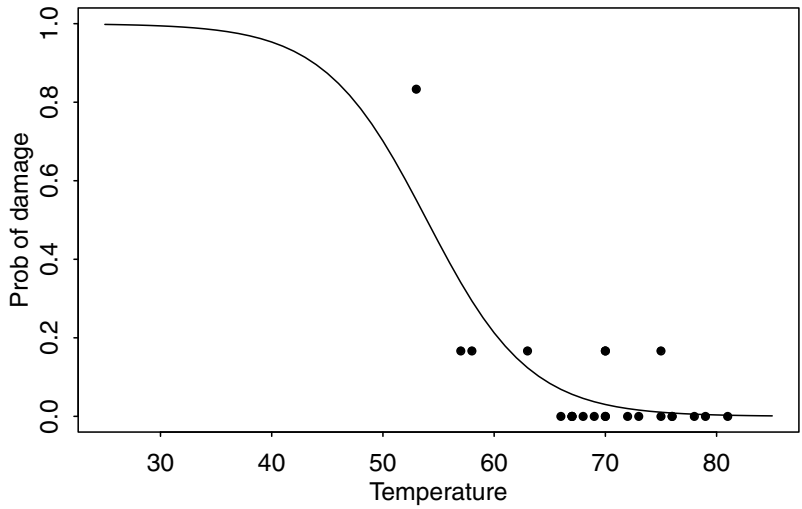


Figure 3.1 *Damage to O-rings in 23 space shuttle missions as a function of launch temperature. Logistic fit is shown.*

We are interested in how the probability of failure in a given O-ring is related to the launch temperature and predicting that probability when the temperature is 31°F. We estimate the regression parameters for the Challenger data:

```
lmod <- glm(cbind(damage,6-damage) ~ temp, family=binomial,orings)
sumary(lmod)
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  11.6630     3.2963    3.54   0.0004
temp         -0.2162     0.0532   -4.07 0.000048

n = 23 p = 2
Deviance = 16.912 Null Deviance = 38.898 (Difference = 21.985)
```

For binomial response data, we need two pieces of information about the response values — y_i and m_i. We express this as a two-column matrix with the first column representing the number of successes y and the second column the number of failures $m - y$. In this case, an O-ring damage incident is a "success."

We show the logit fit to the data as seen in Figure 3.1:

```
x <- seq(25,85,1)
```

```
lines(x,ilogit(11.6630-0.2162*x))
```

We can predict the response at 31°F:

```
ilogit(11.6630-0.2162*31)
```

```
[1] 0.99304
```

We see a very high probability of damage although we still need to develop some inferential techniques before we leap to any conclusion.

3.2 Inference

We use the same likelihood-based methods as in Section 2.3 to derive the binomial deviance:

$$D = 2\sum_{i=1}^{n}\{y_i \log y_i/\hat{y}_i + (m_i - y_i)\log(m_i - y_i)/(m_i - \hat{y}_i)\}$$

where $\hat{y}_i$ are the fitted values from the model.

Provided that Y is truly binomial and that the m_i are relatively large, the deviance is approximately χ^2 distributed with $n-q-1$ degrees of freedom if the model is correct. Thus we can use the deviance to test whether the model is an adequate fit. For the logit model of the Challenger data, we may compute:

```
pchisq(deviance(lmod),df.residual(lmod),lower=FALSE)
```

```
[1] 0.71641
```

Since this p-value is well in excess of 0.05, we conclude that this model fits sufficiently well. Of course, this does not mean that this model is correct or that a simpler model might not also fit adequately. Even so, for the null model:

```
pchisq(38.9,22,lower=FALSE)
```

```
[1] 0.014489
```

We see that the fit is inadequate, so we cannot ascribe the response to simple variation not dependent on any predictor. Note that a χ^2_d variable has mean d and standard deviation $\sqrt{2d}$ so that it is often possible to quickly judge whether a deviance is large or small without explicitly computing the p-value. If the deviance is far in excess of the degrees of freedom, the null hypothesis can be rejected.

The χ^2 distribution is only an approximation that becomes more accurate as the m_i increase. The approximation is very poor for small m_i and fails entirely in binary cases where $m_i = 1$. Although it is not possible to say exactly how large m_i should be for an adequate approximation, $m_i \geq 5\ \forall i$ has often been suggested. Permutation or bootstrap methods might be considered as an alternative.

We can also use the deviance to compare two models, with smaller model S representing a subspace (usually a subset) of a larger model L. The likelihood ratio test statistic becomes $D_S - D_L$. This test statistic is asymptotically distributed χ^2_{l-s}, assuming that the smaller model is correct and the distributional assumptions hold. We can use this to test the significance of temperature by computing the difference in the deviances between the model with and without temperature. The model without temperature is just the null model and the difference in degrees of freedom or parameters is one:

```
pchisq(38.9-16.9,1,lower=FALSE)
```

```
[1] 2.7265e-06
```

Since the p-value is so small, we conclude that the effect of launch temperature is statistically significant. An alternative to this test is the z-value, which is $\hat{\beta}/se(\hat{\beta})$, here equal to -4.07 with a p-value of 4.8e-05. As in the binary case, the deviance-based test is preferred. Again, there are concerns with the accuracy of the approximation, but the test involving differences of deviances is generally more accurate than the goodness of fit test involving a single deviance.

What if all the cases that form a covariate class have not been grouped together? Let's see what happens when each individual O-ring is listed separately:

```
erings <- with(orings, data.frame(temp=rep(temp,each=6), damage=as.
    ↪ vector(sapply(orings$damage, function(x) rep(c(0,1), times=c(6-
    ↪ x,x))))))
```

The first launch was at 53 degrees and resulted in five damaged O-rings out of six. This has been divided into six individual trials:

```
head(erings)
```

```
  temp damage
1   53      0
2   53      1
3   53      1
4   53      1
5   53      1
6   53      1
```

Now we fit the model and examine the ouput:

```
emod <- glm(damage ~ temp, family=binomial, erings)
sumary(emod)
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  11.6630     3.2962    3.54   0.0004
temp         -0.2162     0.0532   -4.07 0.000048

n = 138 p = 2
Deviance = 54.759 Null Deviance = 76.745 (Difference = 21.985)
```

We see that the parameter estimates, standard errors and the difference in the deviances are the same as before so we will make the same conclusions. The only thing we have lost in this version of the output is the ability to make the goodness of fit test from the residual deviance.

Confidence intervals for the regression parameters may be constructed as with the binary regression model:

```
confint(lmod)
```

```
Waiting for profiling to be done...
                2.5 %   97.5 %
(Intercept)  5.57543 18.73812
temp        -0.33267 -0.12018
```

We prefer this profile likelihood method to the $\hat{\beta} \pm 2\mathtt{se}(\hat{\beta})$ approach.

3.3 Pearson's χ^2 Statistic

The deviance is one measure of how well the model fits the data, but there are alternatives. The Pearson's X^2 statistic takes the general form:

$$X^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

where O_i is the observed count and E_i is the expected count for case i. For a binomial response, we count the number of successes for which $O_i = y_i$ while $E_i = n_i\hat{p}_i$ and failures for which $O_i = n_i - y_i$ and $E_i = n_i(1-\hat{p}_i)$, which results in:

$$X^2 = \sum_{i=1}^{n} \frac{(y_i - n_i\hat{p}_i)^2}{n_i\hat{p}_i(1-\hat{p}_i)}$$

If we define *Pearson residuals* as:

$$r_i^P = (y_i - n_i\hat{p}_i)/\sqrt{\text{var}\,\hat{y}_i}$$

which can be viewed as a type of standardized residual, then $X^2 = \sum_{i=1}^{n}(r_i^P)^2$. So the Pearson's X^2 is analogous to the residual sum of squares used in normal linear models.

The Pearson X^2 will typically be close in size to the deviance and can be used in the same manner. Alternative versions of the hypothesis tests described above might use the X^2 in place of the deviance with the same approximate null distributions. However, some care is necessary because the model is fit to minimize the deviance and not the Pearson's X^2. This means that it is possible, although unlikely, that the X^2 could increase as a predictor is added to the model. X^2 can be computed like this:

```
[1] 28.067
```

Compare this to:

```
deviance(lmod)
```

```
[1] 16.912
```

In this case there is more than the typical small difference between X^2 and the deviance. However, a test for model fit:

```
1-pchisq(28.067, 21)
```

```
[1] 0.13826
```

results in a moderate sized p-value which would not reject this model which agrees with decision based on the deviance statistic.

3.4 Overdispersion

If the binomial model specification is correct, we expect that the residual deviance will be approximately distributed χ^2 with the appropriate degrees of freedom. Sometimes, we observe a deviance that is much larger than would be expected if the model were correct. We must then determine which aspect of the model specification is incorrect.

The most common explanation is that we have the wrong structural form for the model. We have not included the right predictors or we have not transformed or combined them in the correct way. We have a number of ways of determining the importance of potential additional predictors and diagnostics for determining better transformations — see Section 8.4. Suppose, however, that we are able to exclude this explanation. This is difficult to achieve, but when we have only one or two predictors, it is feasible to explore the model space quite thoroughly and be sure that there is not a plausible superior model formula.

Another common explanation for a large deviance is the presence of a small

number of outliers. Fortunately, these are easily checked using diagnostic methods. When larger numbers of points are identified as outliers, they become unexceptional, and we might more reasonably conclude that there is something amiss with the error distribution.

Sparse data can also lead to large deviances. In the extreme case of a binary response, the deviance is not even approximately χ^2. In situations where the group sizes are simply small, the approximation is poor. Because we cannot judge the fit using the deviance, we shall exclude this case from further consideration in this section.

Having excluded these other possibilities, we might explain a large deviance by deficiencies in the random part of the model. A binomial distribution for Y arises when the probability of success p is independent and identical for each trial within the group. If the group size is m, then $\text{var}\, Y = mp(1-p)$ if the binomial assumptions are correct. However, if the assumptions are broken, the variance may be greater. This is *overdispersion*. In rarer cases, the variance is less and *underdispersion* results.

There are two main ways that overdispersion can arise — the independent or identical assumptions can be violated. We look at the constant p assumption first. It is easy to see how there may be some unexplained heterogeneity within a group that might lead to some variation in p. For example, in the shuttle disaster case study of Section 3.1, the position of the O-ring on the booster rocket may have some effect on the failure probability. Yet this variable was not recorded and so we cannot include it as a predictor. Heterogeneity can also result from clustering. Suppose a population is divided into clusters, so that when you take a sample, you actually get a sample of clusters. This would be common in epidemiological applications.

Let the sample size be m, the cluster size be k and the number of clusters be $l = m/k$. Let the number of successes in cluster i be $Z_i \sim B(k, p_i)$. Now suppose that p_i is a random variable such that $Ep_i = p$ and $\text{var}\, p_i = \tau^2 p(1-p)$. Let the total number of successes be $Y = Z_1 + \cdots + Z_l$. Then:

$$EY = \sum EZ_i = \sum_{i=1}^{l} kp = mp$$

as in the standard case, but:

$$\text{var}\, Y = \sum \text{var}\, Z_i = \sum \{E(\text{var}\,(Z_i|p_i)) + \text{var}\,(E(Z_i|p_i))\} = (1+(k-1)\tau^2)mp(1-p)$$

So Y is overdispersed since $1+(k-1)\tau^2 \geq 1$. Notice that in the sparse case, $m = 1$, and this problem cannot arise.

Overdispersion can also result from dependence between trials. If the response has a common cause, say a disease is influenced by genes, the responses will tend to be positively correlated. For example, subjects in human or animal trials may be influenced in their responses by other subjects. If the food supply is limited, the probability of survival of an animal may be increased by the death of others. This circumstance would result in underdispersion.

The simplest approach for modeling overdispersion is to introduce an additional dispersion parameter, so that $\texttt{var}Y = \sigma^2 mp(1-p)$. In the standard binomial case

$\sigma^2 = 1$. We now let σ^2 vary and estimate using the data. Notice the similarity to linear regression. The dispersion parameter may be estimated using:

$$\hat{\sigma}^2 = \frac{X^2}{n-p}$$

Using the deviance in place of the Pearson's X^2 is not recommended as it may not be consistent. The estimation of β is unaffected since σ^2 does not change the mean response but:

$$\hat{\text{var}}\,\hat{\beta} = \hat{\sigma}^2 (X^T \hat{W} X)^{-1}$$

So we need to scale up the standard errors by a factor of $\hat{\sigma}$.

We cannot use the difference in deviances when comparing models, because the test statistic will be distributed $\sigma^2\chi^2$. Since σ^2 is not known and must be estimated in the overdispersion situation, an F-statistic must be used:

$$F = \frac{(D_{small} - D_{large})/(df_{small} - df_{large})}{\hat{\sigma}^2}$$

This statistic is only approximately F distributed, in contrast to the Gaussian case.

In Manly (1978), an experiment is reported where boxes of trout eggs were buried at five different stream locations and retrieved at four different times, specified by the number of weeks after the original placement. The number of surviving eggs was recorded. The box was not returned to the stream. The data is also analyzed by Hinde and Demetrio (1988). We can construct a tabulation of the data by:

```
data(troutegg, package="faraway")
ftable(xtabs(cbind(survive,total) ~ location+period, troutegg))
```

```
                survive total
location period
1        4           89    94
         7           94    98
         8           77    86
         11         141   155
2        4          106   108
         7           91   106
         8           87    96
         11         104   122
3        4          119   123
         7          100   130
         8           88   119
         11          91   125
4        4          104   104
         7           80    97
         8           67    99
         11         111   132
5        4           49    93
         7           11   113
         8           18    88
         11           0   138
```

Notice that in one case, all the eggs survive, while in another, none of the eggs survive. We now fit a binomial GLM for the two main effects:

```
bmod <- glm(cbind(survive,total-survive) ~ location+period, family=
    ↪ binomial,troutegg)
sumary(bmod)
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)    4.636      0.281   16.48  < 2e-16
location2     -0.417      0.246   -1.69     0.09
location3     -1.242      0.219   -5.66  1.5e-08
location4     -0.951      0.229   -4.16  3.2e-05
location5     -4.614      0.250  -18.44  < 2e-16
period7       -2.170      0.238   -9.10  < 2e-16
period8       -2.326      0.243   -9.57  < 2e-16
period11      -2.450      0.234  -10.47  < 2e-16

n = 20 p = 8
Deviance = 64.495 Null Deviance = 1021.469 (Difference = 956.974)
```

The deviance of 64.5 on 12 degrees of freedom seems to show that this model does not fit. Before we conclude that there is overdispersion, we need to eliminate other potential explanations. With about 100 eggs in each box, we have no problem with sparseness, but we do need to check for outliers and look at the model formula. A half-normal plot of the residuals is a good way to check for outliers:

```
halfnorm(residuals(bmod))
```

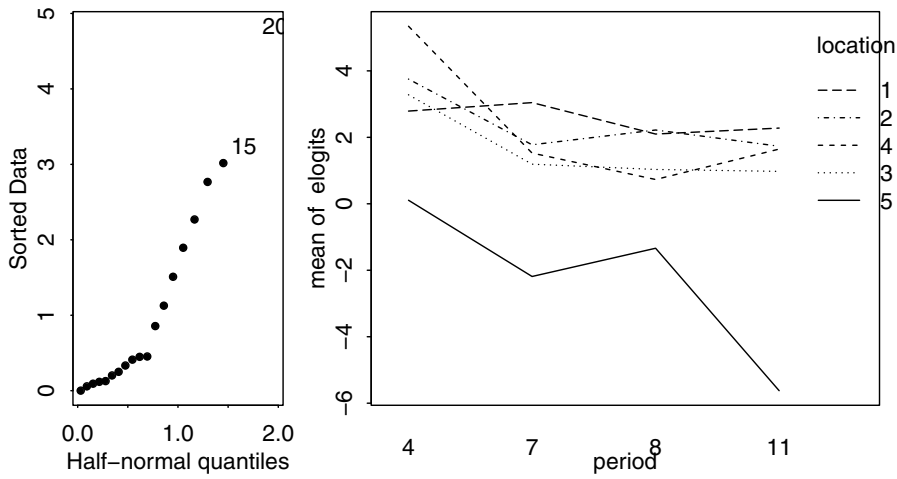


Figure 3.2 *Diagnostic plots for the trout egg model. A half-normal plot of the residuals is shown on the left and an interaction plot of the empirical logits is shown on the right.*

The half-normal plot is shown in the left panel of Figure 3.2. No single outlier is apparent. Perhaps one can discern a larger number of residuals which seem to follow a more dispersed distribution than the rest.

We can also check whether the predictors are correctly expressed by plotting the *empirical logits*. These are defined as:

$$\log\left(\frac{y+1/2}{m-y+1/2}\right)$$

The halves are added to prevent infinite values for groups consisting of all successes or failures. We now construct an interaction plot of the empirical logits:

```
elogits <- with(troutegg,log((survive+0.5)/(total- survive+0.5)))
with(troutegg,interaction.plot(period,location,elogits))
```

Interaction plots are difficult to interpret conclusively, but there is no obvious sign of large interactions. So there is no evidence that the linear model is inadequate. We do not have any outliers and the functional form of the model appears to be suitable, but the deviance is still larger than should be expected. Having eliminated these more obvious causes as the source of the problem, we may now put the blame on overdispersion. Possible reasons for the overdispersion include inhomogeneous trout eggs, variation in the experimental procedures or unknown variables affecting survival.

We can estimate the dispersion parameter as:

```
(sigma2 <- sum(residuals(bmod,type="pearson")^2)/12)
```

```
[1] 5.3303
```

We see that this is substantially larger than one as it would be in the standard binomial GLM. We can now make F-tests on the predictors using:

```
drop1(bmod,scale=sigma2,test="F")
```

```
Single term deletions
scale:  5.3303

         Df Deviance AIC F value   Pr(F)
<none>            64 157
location  4      914 308    39.5 8.1e-07
period    3      229 182    10.2  0.0013
Warning message:
F test assumes quasibinomial family in:
 drop1.glm(bmod, scale = sigma2, test = "F")
```

We see that both terms are clearly significant. It is necessary to specify the `scale` argument using the estimated value of σ^2. If this argument is omitted, the deviance will be used in the estimation of the dispersion parameter. For this particular dataset, it makes very little difference, but in some cases, using the deviance to estimate the dispersion gives inconsistent results. The warning message reminds us that the use of free dispersion parameter results in a model that is no longer a true binomial GLM, but rather what is known as a *quasi-binomial* GLM. More on such models may be found in Section 3.5.

No goodness of fit test is possible because we have a free dispersion parameter. We can use the dispersion parameter to scale up the estimates of the standard error as in:

```
sumary(bmod,dispersion=sigma2)
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)    4.636      0.649    7.14  9.5e-13
location2     -0.417      0.568   -0.73    0.463
location3     -1.242      0.507   -2.45    0.014
location4     -0.951      0.528   -1.80    0.072
location5     -4.614      0.578   -7.99  1.4e-15
period7       -2.170      0.550   -3.94  8.1e-05
period8       -2.326      0.561   -4.15  3.4e-05
period11      -2.450      0.540   -4.53  5.8e-06

overdispersion parameter = 5.330
```

```
n = 20 p = 8
Deviance = 64.495 Null Deviance = 1021.469 (Difference = 956.974)
```

We see that the differences in the location become less pronounced with only the fifth location being clearly different.

This dispersion parameter method is only appropriate when the covariate classes are roughly equal in size. If not, more sophisticated methods should be used. One such approach uses the beta-binomial distribution where we assume that p follows a beta distribution. This approach is discussed in Williams (1982) and Crowder (1978) and can be implemented using the `dispmod` package of Scrucca (2012) in R:

```
library(dispmod)
dmod <- glm.binomial.disp(bmod)
summary(dmod)
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)    4.518      0.621    7.28  3.3e-13
location2     -0.377      0.560   -0.67    0.501
location3     -1.210      0.507   -2.39    0.017
location4     -0.956      0.520   -1.84    0.066
location5     -4.468      0.559   -8.00  1.3e-15
period7       -2.086      0.520   -4.01  6.1e-05
period8       -2.227      0.523   -4.26  2.0e-05
period11      -2.362      0.519   -4.56  5.2e-06

n = 20 p = 8
Deviance = 12.400 Null Deviance = 190.186 (Difference = 177.786)
```

As can be seen, the results are quite similar to the previous approach because the covariate class sizes do not vary much.

3.5 Quasi-Binomial

In the previous section, we have demonstrated ways to model data where the supposedly binomial response is more variable than should be expected. A *quasi-binomial* model is another way to allow for extra-binomial variation. We will explain the method in greater generality than immediately necessary because the idea can be used across a wider range of response types.

The idea is to specify only how the mean and variance of the response are connected to the linear predictor. The method of weighted least squares, as used for standard linear models, would be a simple example of this. An examination of the fitting of the binomial model reveals that this only requires the mean and variance information and does not use any additional information about the binomial distribution. Hence, we can obtain the parameter estimates $\hat{\beta}$ and standard errors without making the full binomial assumption.

The problem arises when we attempt to do inference. To construct a confidence interval or perform an hypothesis test, we need some distributional assumptions. Previously we have used the deviance, but for this we need a likelihood and to compute a likelihood we need a distribution. Now we need a suitable substitute for a likelihood that can be computed without assuming a distribution.

Let Y_i have mean μ_i and variance $\phi V(\mu_i)$. We assume that Y_i are independent. We

define a score, U_i:

$$U_i = \frac{Y_i - \mu_i}{\phi V(\mu_i)}$$

Now:

$$EU_i = 0$$

$$\text{var } U_i = \frac{1}{\phi V(\mu_i)}$$

$$-E\frac{\partial U_i}{\partial \mu_i} = -E\frac{-\phi V(\mu_i) - (Y_i - \mu_i)\phi V'(\mu_i)}{[\phi V(\mu_i)]^2} = \frac{1}{\phi V(\mu_i)}$$

These properties are shared by the derivative of the log-likelihood, l'. This suggests that we can use U in place of l'. So we define:

$$Q_i = \int_{y_i}^{\mu_i} \frac{y_i - t}{\phi V(t)} dt$$

The intent is that Q should behave like the log-likelihood. We then define the log *quasi-likelihood* for all n observations as:

$$Q = \sum_{i=1}^{n} Q_i$$

The usual asymptotic properties expected of maximum likelihood estimators also hold for quasi-likelihood-based estimators as may be seen in McCullagh (1983).

Notice that the quasi-likelihood depends directly only on the variance function and that the choice of distribution also determines only the variance function. So the choice of variance function is associated with the random structure of the model while the link function determines the relationship with the systematic part of the model.

For the standard linear model, the quasi-likelihood corresponds exactly to the log-likelihood. Here the *dispersion parameter* ϕ is σ^2 so nothing is gained by this approach. However, for the binomial model, the introduction of ϕ provides an additional dimension of flexibility to the model, which is useful in modeling overdispersion. One curious possibility is that some choices of $V(\mu)$ may not correspond to a known, or even any, distribution.

$\hat{\beta}$ is obtained by maximizing Q. Everything proceeds as before except for the estimation of ϕ since the likelihood approach is not reliable here. We recommend:

$$\hat{\phi} = \frac{X^2}{n-p}$$

Although quasi-likelihood estimators are attractive because they require fewer assumptions, they are generally less efficient than the corresponding regular likelihood-based estimator. So if you have information about the distribution, you are advised to use it.

The inferential procedures are similar to those used before. Recall that the regular

deviance for a model is formed from the difference in log-likelihoods for the model and the saturated model:

$$D(y,\hat{\mu}) = -2\phi\sum_i (l(\hat{\mu}_i|y_i) - l(y_i|y_i))$$

so by analogy the quasi-deviance is $-2\phi Q$ because the contribution from the saturated model is zero. The ϕ cancels, so the quasi-deviance is just:

$$Q = -2\sum_i \int_{y_i}^{\mu_i} \frac{y_i - t}{V(t)} dt$$

In Allison and Cicchetti (1976), data on the sleep behavior of 62 mammals is presented. Suppose we are interested in modeling the proportion of sleep spent dreaming as a function of the other predictors: the weight of the body and the brain, the gestation period, the lifespan and the three constructed indices measuring vulnerability to predation, exposure while sleeping and overall danger:

```
data(mammalsleep, package="faraway")
mammalsleep$pdr <- with(mammalsleep, dream/sleep)
summary(mammalsleep$pdr)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  0.000   0.118   0.176   0.186   0.243   0.462  14.000
```

We notice that the proportion of time spent dreaming varies from zero up to almost half the time. For some datasets with a proportion response, the range of proportions might never come close to zero or one. For such datasets, using a normal Gaussian model (with perhaps some weights) might be acceptable. But for this dataset, we have some very small proportions as response values and so a Gaussian model will not work. We attempt to model the proportion response directly. A logit link seems sensible since the response is restricted between zero and one. Furthermore, we might expect the variance to be greater for moderate values of the proportion μ and less as μ approaches zero or one because of the nature of the measurements. This suggests a variance function of the approximate form $\mu(1-\mu)$. This corresponds to the binomial GLM with the canonical logit link and yet the response is not binomial. We can use a quasi-binomial:

```
modl <- glm(pdr ~ log(body)+ log(brain) + log(lifespan) + log(
    ↪ gestation) + predation + exposure + danger, family=
    ↪ quasibinomial, mammalsleep)
```

where we have logged many of the predictors because of skewness. Since we now have a free dispersion parameter, we must use F-tests to compare models:

```
drop1(modl,test="F")
```

```
Single term deletions
              Df Deviance F value Pr(F)
<none>              1.57
log(body)      1     1.78    4.51 0.041
log(brain)     1     1.59    0.33 0.568
log(lifespan)  1     1.65    1.79 0.189
log(gestation) 1     1.62    1.15 0.292
predation      1     1.57    0.10 0.749
exposure       1     1.58    0.32 0.575
danger         1     1.58    0.31 0.579
```

We might eliminate `predation` as the least significant variable. Further sequential backward elimination results in:

```
modl <- glm(pdr ~ log(body) + log(lifespan) + danger, family=
    ↪ quasibinomial, mammalsleep )
sumary(modl)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.4932     0.2913   -1.69  0.09796
log(body)      0.1463     0.0384    3.81  0.00046
log(lifespan) -0.2866     0.1080   -2.65  0.01126
danger        -0.1732     0.0600   -2.89  0.00615

overdispersion parameter = 0.041
n = 45 p = 4
Deviance = 1.732 Null Deviance = 2.509 (Difference = 0.777)
```

Notice that the dispersion parameter is far less than the default value of one that we would see for a binomial. We see the proportion of time spent dreaming increases for heavier mammals that live less time and live in less danger. Notice that the relatively large residual deviance compared to the null deviance indicates that this is not a particularly well-fitting model.

The usual diagnostics should be performed. Here are two selected plots that have some interest:

```
ll <- row.names(na.omit(mammalsleep[,c(1,6,10,11)]))
halfnorm(cooks.distance(modl),labs=ll)
plot(predict(modl),residuals(modl,type="pearson"),  xlab="Linear
    ↪ Predictor", ylab="Pearson Residuals")
```

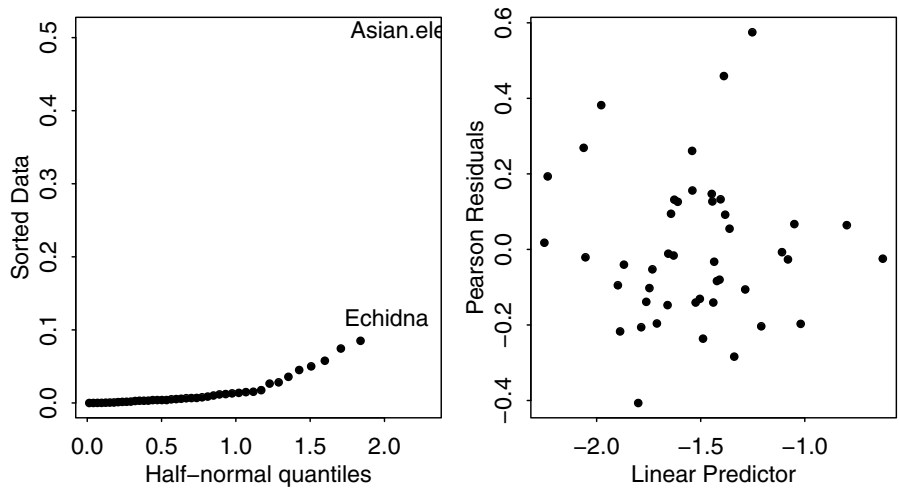


Figure 3.3 *A half-normal plot of the Cook statistics is shown on the left and a plot of the Pearson residuals against the fitted linear predictors is shown on the right.*

In the first panel of Figure 3.3, we see that the Asian elephant is quite influential and a fit without this case should be considered. In the second panel, we see that a pattern of constant variation indicating that our choice of variance function was reasonable. We used the Pearson residuals because these explicitly normalize the raw residuals

using the variance function making the check more transparent. Even so, the deviance residuals would have served the same purpose.

3.6 Beta Regression

Beta regression is useful for responses that are bounded in $(0,1)$ such as proportions. It could also be used for variables that are bounded in some other finite interval simply by rescaling to $(0,1)$. A Beta-distributed random variable Y has density:

$$f(y|a,b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} y^{a-1}(1-y)^{b-1}$$

for parameters a,b and Gamma function $\Gamma()$. It is more convenient to transform the parameters so $\mu = a/(a+b)$ and $\phi = a+b$ so that $EY = \mu$ and $\text{var}\, Y = \mu(1-\mu)/(1+\phi)$. We can then link the linear predictor η using $\eta = g(\mu)$ using a link function g where any of the choices used for the binomial model would be suitable.

An implementation of the Beta regression model can be found in the `mgcv` package of Wood (2006) . We can apply this to the `mammalsleep` also used in the previous section.

```
data(mammalsleep, package="faraway")
mammalsleep$pdr <- with(mammalsleep, dream/sleep)
library(mgcv)
modb <- gam(pdr ~ log(body)+log(lifespan), family=betar(), mammalsleep
    ↪ )
summary(modb)
```

```
Family: Beta regression(8.927)
Link function: logit

Formula:
pdr ~ log(body) + log(lifespan)

Parametric coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)     0.3779     0.3732    1.01     0.31
log(body)       0.2680     0.0551    4.86  1.2e-06
log(lifespan)  -0.9227     0.1658   -5.56  2.6e-08


R-sq.(adj) =  -0.178   Deviance explained = 73.5%
-REML = -47.801  Scale est. = 1         n = 45
```

The default choice of link is the logit function. The estimated value of ϕ is 8.927. A comparison of the fitted values of this model and the quasi-binomial model fitted earlier reveals no substantial difference. The advantage of the Beta-based model is the full distributional model which would allow the construction of full predictive distributions rather than just a point estimate and standard error.

Exercises

1. The question concerns data from a case-control study of esophageal cancer in Ille-et-Vilaine, France. The data is distributed with R and may be obtained along with a description of the variables by:

```
data(esoph)
help(esoph)
```

(a) Plot the proportion of cases against each predictor using the size of the point to indicate the number of subject as seen in Figure 2.7. Comment on the relationships seen in the plots.

(b) Fit a binomial GLM with interactions between all three predictors. Use AIC as a criterion to select a model using the `step` function. Which model is selected?

(c) All three factors are ordered and so special contrasts have been used appropriate for ordered factors involving linear, quadratic and cubic terms. Further simplification of the model may be possible by eliminating some of these terms. Use the `unclass` function to convert the factors to a numerical representation and check whether the model may be simplified.

(d) Use the summary output of the factor model to suggest a model that is slightly more complex than the linear model proposed in the previous question.

(e) Does your final model fit the data? Is the test you make accurate for this data?

(f) Check for outliers in your final model.

(g) What is the predicted effect of moving one category higher in alcohol consumption?

(h) Compute a 95% confidence interval for this predicted effect.

2. Incubation temperature can affect the sex of turtles. An experiment was conducted with three independent replicates for each temperature and the number of male and female turtles born was recorded and can be found in the `turtle` dataset.

(a) Plot the proportion of males against the temperature. Comment on the nature of the relationship.

(b) Fit a binomial response model with a linear term in temperature. Does this model fit the data?

(c) Is this data sparse?

(d) Check for outliers.

(e) Compute the empirical logits and plot these against temperature. Does this indicate a lack of fit?

(f) Add a quadratic term in temperature. Is this additional term a significant predictor of the response. Does the quadratic model fit the data?

(g) There are three replicates for each value of temperature. Assuming independent binomial variation, how much variation would be expected in the three proportions observed? Compare this to the observed variation in these proportions. Do they approximately agree or is there evidence of greater variation?

(h) If the three replicates are homogenous, they could be combined so that the dataset would have only five cases in total. Create this dataset and fit a model linear in temperature. Compare the fit seen for this model with that found in (b).

3. A biologist analyzed an experiment to determine the effect of moisture content on seed germination. Eight boxes of 100 seeds each were treated with the same moisture level. Four boxes were covered and four left uncovered. The process was repeated at six different moisture levels.
   (a) Plot the germination percentage against the moisture level on two side-by-side plots according to the coverage of the box. What relationship do you see?
   (b) Create a new factor describing the box (the data are ordered in blocks of 6 observations per box). Add lines to your previous plot that connect observations from the same box. Is there an indication of a box effect?
   (c) Fit a binomial response model including the coverage, box and moisture predictors. Use the plots to determine an appropriate choice of model.
   (d) Test for the significance of a box effect in your model. Repeat the same test but using the Pearson's Chi-squared statistic instead of the deviance.
   (e) At what value of moisture does the predicted maximum germination occur for noncovered boxes? For covered boxes?
   (f) Produce a plot of the residuals against the fitted values and interpret.
   (g) Plot the residuals against moisture while distinguishing the covering. Interpret.
   (h) Plot the residuals against the leverages. Are there any influential points?
4. This problem concerns the modeling of the quantitative structure-activity relationships (QSAR) of the inhibition of dihydrofolate reductase (DHFR) by pyrimidines. We want to relate the physicochemical and/or structural properties as exhibited by the 26 predictors in `pyrimidines` with an activity level. We have structural information on 74 2,4-diamino- 5-(substituted benzyl) pyrimidines used as inhibitors of DHFR in *E. coli*. All the variables lie in [0,1].
   (a) Plot the activity (response) against the first three predictors. Are any outliers in the response apparent? Remove any such cases.
   (b) Fit a Gaussian linear model for the response with all 26 predictors. How well does this model fit the data in terms of R^2? Plot the residuals against the fitted values. Is there any evidence of a violation of the standard assumptions?
   (c) Fit a quasi-binomial model for the activity response. Compare the predicted values for this model to those for the Gaussian linear model. Take care to compute the predicted values in the appropriate scale. Compare the fitted coefficients between the two models. Are there any substantial differences?
   (d) Fit a Gaussian linear model with the logit transformation applied to the response. Compare the coefficients of this model with the quasi-binomial model.
   (e) Fit a Beta regression model. Compare the coefficients of this model with that of logit response regression model.
   (f) What property of the response leads to the similarity of the models considered thus far in this question?